SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	**Assessment:**	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO2 Statement:	*Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight.(BL4)*		
Student Name:	P.Vishwa nandan	**Reg. No.:**	192473015
Section / Batch:	2024	**Date:**	30-07-2026

Code of Conduct

I P.Vishwa nandan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: vishwa

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Development Exercise

CO2 · BT4: Analyze ·

This rubric is used to assess student responses to the CO2-AT2 Visualization Development Exercise, in which students build a small multi-chart dashboard from given data, add at least one annotation, and write a narrative takeaway.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of which chart types to combine, how they relate to each other, and how annotation and narrative should be used to communicate insight.	Good understanding of chart combination and annotation/narrative technique, with only minor gaps.	Basic understanding of dashboard development, but with some misconceptions about which charts to combine or how to annotate them.	Poor understanding of core development concepts; combines charts or annotations that do not suit the data or the story being told.
Explanation	**25%**	The annotation and narrative takeaway are precise, well-justified, and clearly tied to a specific, correctly identified data point or pattern.	The annotation and narrative are clear and reasonably well-justified, with adequate connection to the data.	The annotation or narrative is present but vague, generic, or weakly connected to the actual data shown.	Annotation or narrative is missing, incorrect, or unconnected to the data in the dashboard.
Application	**20%**	Effectively applies multi-chart coordination, annotation, and storytelling techniques to produce a	Applies development techniques to produce a workable dashboard, with only minor errors or missed refinements.	Limited application; the dashboard partially addresses the scenario but charts feel disconnected or the story is unclear.	Fails to apply appropriate development techniques; the charts, annotation, and narrative do not work together as

		complete, cohesive dashboard for the scenario.			a dashboard.
Organization	**15%**	Charts are clearly coordinated (aligned axes or shared theme), laid out logically, with the annotation and narrative placed for maximum clarity.	Charts are organized and mostly coordinated, with only minor issues in alignment or layout.	Basic structure; charts are present but poorly aligned or the annotation/narrative feels disconnected from the layout.	Poorly organized; charts are not coordinated, and the annotation or narrative is missing or misplaced.
Accuracy	**10%**	All chart values, scales, and the annotated data point are completely accurate and correctly reflect the given data.	Mostly accurate, with only minor omissions or a small error in one chart or the annotation.	Partially correct; some plotted values or the annotated point are missing or incorrect.	Largely inaccurate; major errors in plotted values, scales, or the annotated data point.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Name:P.Vishwa Nandan
Reg no:192473015
Sub:Data handling and Visualization
Sub code:DSA0606

CO2 — AT2: VISUALIZATION DEVELOPMENT EXERCISE

Problem 11 – Sports Team Performance Dashboard

Code

```
# Data
teams <- c("A","B","C","D")
wins <- c(18,14,20,10)
losses <- c(6,10,4,14)
points <- c(2200,2100,2400,1900)

# Dashboard Layout
par(mfrow=c(1,2))

# Bar Chart 1 - Wins
barplot(wins,
    names.arg=teams,
    col="lightblue",
    main="Wins by Team",
    ylab="Wins")

text(3,20,"Best Team",pos=3)

# Bar Chart 2 - Points
barplot(points,
    names.arg=teams,
```

```
        col="lightgreen",
        main="Points Scored",
        ylab="Points")

text(3,2400,"Highest Points",pos=3)
```

Problem 12 – Restaurant Chain Dashboard

Code

```
months <- c("Jan","Feb","Mar","Apr")

downtown <- c(40,45,42,50)
uptown <- c(30,32,28,25)
suburb <- c(35,38,40,44)

satisfaction <- c(4.5,3.2,4.1)

par(mfrow=c(1,2))

# Line Chart
plot(downtown,type="o",
     col="blue",
     ylim=c(20,55),
     xaxt="n",
     xlab="Month",
     ylab="Revenue",
     main="Monthly Revenue")

axis(1,at=1:4,labels=months)
```

```
lines(uptown,type="o",col="red")
lines(suburb,type="o",col="green")

legend("topleft",
legend=c("Downtown","Uptown","Suburb"),
col=c("blue","red","green"),
lty=1)

text(4,25,"Declining Revenue",pos=3)

# Satisfaction
barplot(satisfaction,
	names.arg=c("Downtown","Uptown","Suburb"),
	col="orange",
	main="Average Satisfaction",
	ylab="Rating")
```

Narrative

Uptown has the lowest customer satisfaction and shows declining revenue over time.

Problem 13 – App Usage Dashboard

Code

```
days <- c("Mon","Tue","Wed","Thu","Fri","Sat","Sun")

users <- c(2000,2200,2100,2500,3000,3500,3200)
duration <- c(5,4.5,6,5.5,4,7,6.5)

par(mfrow=c(1,2))

# Bar Chart
```

```
barplot(users,
        names.arg=days,
        col="skyblue",
        main="Daily Active Users",
        ylab="Users")

text(6,3500,"Highest Users",pos=3)

# Line Chart
plot(duration,
     type="o",
     xaxt="n",
     xlab="Day",
     ylab="Minutes",
     main="Session Duration")

axis(1,at=1:7,labels=days)

text(6,7,"Highest Engagement",pos=3)
```

Problem 14 – Supply Chain Dashboard

Code

```
carrier <- c("A","B","C","D")

volume <- c(500,700,300,900)
delivery <- c(95,80,99,70)

par(mfrow=c(1,2))
```

```
# Shipment Volume
barplot(volume,
        names.arg=carrier,
        col="lightblue",
        main="Shipment Volume",
        ylab="Shipments")

# Delivery Rate
barplot(delivery,
        names.arg=carrier,
        col="pink",
        main="On-Time Delivery",
        ylab="Percent")

text(4,70,"Risk",pos=3)
```

Problem 15 – University Admissions Dashboard

Code

```
years <- c(1,2,3)

engineering <- c(12,15,20)
business <- c(18,17,16)
arts <- c(8,9,7)

acceptance <- c(25,40,55)

par(mfrow=c(1,2))

# Multi-line Chart
```

```
plot(years,engineering,
     type="o",
     col="blue",
     ylim=c(5,22),
     xaxt="n",
     xlab="Year",
     ylab="Applications (Hundreds)",
     main="Applications Trend")

axis(1,at=1:3,labels=c("Year1","Year2","Year3"))

lines(years,business,type="o",col="red")
lines(years,arts,type="o",col="green")

legend("topleft",
legend=c("Engineering","Business","Arts"),
col=c("blue","red","green"),
lty=1)

text(3,20,"Most Applications",pos=3)

# Acceptance Rate
barplot(acceptance,
        names.arg=c("Engineering","Business","Arts"),
        col="gold",
        main="Acceptance Rate",
        ylab="Percent")

text(1,25,"Most Competitive",pos=3)
```